CT-based Patient Triage of COVID-19 - Prediction of ICU Admission and Death (lab-data adaptation)

UE24CS352A Machine Learning - Mini-Project | Group 23, Problem Statement 12 | Prajwal S Jyanopanthar (PES1UG24CS594), Prakash N Heroor (PES1UG24CS595)

1. Problem statement

Problem 12 reproduces Zhan & Li (CS229, 2020): predict, at hospital admission, which COVID-19 inpatients will be admitted to the ICU, need mechanical ventilation, or die, so scarce resources can go to high-risk patients. The original used CT radiomics + clinical + lab data from a private cohort (1662 + 700 patients, Jinling Hospital); neither the data nor the authors' code repository is public. We therefore adapt the study to a **public lab-only dataset**: the same data-fusion/imbalance methodology, two of the three outcomes (ICU admission, death), no radiomics.

2. Dataset

Routine Blood Values Datasets (Huyut, Mendeley Data, v1, DOI 10.17632/w9mdp9k472.1; non-commercial/educational use). Each task is split into biochemical, hematological and immunological files. **ICU task:** 2000 adults (279 ICU, 14.0%), 54 features (25 biochemical, 17 hematological, 10 immunological, age, sex). **Death task:** 2597 adults (233 died, 9.0%), 37 features (12 + 15 + 10), no age/sex. There is no patient ID, so the three files are joined by row position; this is justified because label order is identical across files (death) and labels agree wherever two files both carry one (ICU), but it remains an assumption. The hematology/immunology ICU files leave 1035 labels blank, so labels come from the biochemical file.

Data-quality findings (important). (i) Missing lab values were filled with constants; in the death files the constant differs by class (e.g. Ferritin = 395 for 92% of deaths, D-dimer = 1277 for 80%), which leaks the label (raw single-block AUROC = 1.00). We blank any value covering >25% of one class in a column (or whose most common value differs between classes) and treat impossible zeros as missing: 17.0k (ICU) and 32.8k (death) values. (ii) Even then, *which* values are missing predicts the outcome: a model given only the missingness mask scores AUROC 0.87 (ICU) / 0.97 (death). Median imputation would let tree models read this mask, so we impute with random draws from each feature's observed training values. Residual leakage cannot be excluded; the hematological block (no repeated rows) is our least-affected estimate.

3. Approach

Following the paper: **(a) imbalance:** SMOTEENN vs. none; **(b) feature selection:** LASSO (L1 logistic), PCA (95% variance), FPR test, or none; **(c) classifiers:** Logistic Regression, Random Forest, SVM (RBF), MLP, LightGBM. All 40 combinations were scored per task with 5-fold stratified CV on a stratified 70% training split (preprocessing, resampling and selection are refit inside each fold, so nothing leaks). The configuration with the best CV AUROC was refit on the 70% and evaluated **once** on the untouched 30% test set. As in the paper's Radiom/Clin/Lab comparison, the winning configuration is re-trained on each data type (biochemical, hematological, immunological, all) and compared with a plain logistic-regression baseline using a paired bootstrap of the test set (200 resamples; 95% percentile CIs). The decision threshold is Youden's J on out-of-fold training scores. Importance is permutation importance on the test set.

4. Implementation

Python 3 / scikit-learn, imbalanced-learn, LightGBM. `src/data.py` loads, aligns and cleans the .sav files; `pipeline.py` defines the imputer, winsorizer (1-99%), scaler, SMOTEENN, selectors and models; `run.py` runs CV, test, bootstrap, importance and saves models; `plots.py`, `demo.py` and `build_report.py` make figures, the live demo and this PDF. Run time is about two minutes on 2 CPU cores; the seed is fixed (42).

5. Results (held-out 30% test set)

Data used	ICU AUROC (95% CI)	ICU AUPRC	Death AUROC (95% CI)	Death AUPRC
All lab features	**0.90 (0.87-0.93)**	**0.64**	**0.94 (0.91-0.96)**	**0.71**
Biochemical only	0.88 (0.84-0.92)	0.63	0.83 (0.79-0.89)	0.46
Hematological only	0.51 (0.45-0.57)	0.14	0.83 (0.78-0.87)	0.36
Immunological only	0.55 (0.50-0.61)	0.16	0.73 (0.66-0.79)	0.37
Baseline LR (all)	0.82 (0.76-0.87)	0.50	0.86 (0.79-0.90)	0.57

Best models: ICU - LightGBM with FPR test (F-classif) and no resampling (CV AUROC 0.91); death - LightGBM with no selection and no resampling (CV AUROC 0.96). At the chosen threshold the ICU model reaches sensitivity 0.79 / specificity 0.83 / precision 0.43 (66 TP, 87 FP, 18 FN, 429 TN); the death model 0.80 / 0.90 / 0.44 (56 TP, 70 FP, 14 FN, 640 TN). Using all lab blocks beat every alternative in 100% or more of the death bootstrap resamples (AUROC); for ICU the margin over the biochemical block alone is smaller (98% of resamples), because biochemistry carries nearly all the ICU signal (hematological 0.51, immunological 0.55, i.e. near chance). SMOTEENN did not help here: best CV AUROC/AUPRC with it is 0.87/0.56 (ICU) and 0.95/0.76 (death), versus 0.91/0.67 and 0.96/0.79 without it, unlike in the paper (threshold-free metrics; we tune the decision threshold instead). Top features: ICU - ALT, Calcium, Age, AST, GGT; death - BASO, eGFR, RDW, Troponin, Procalcitonin.

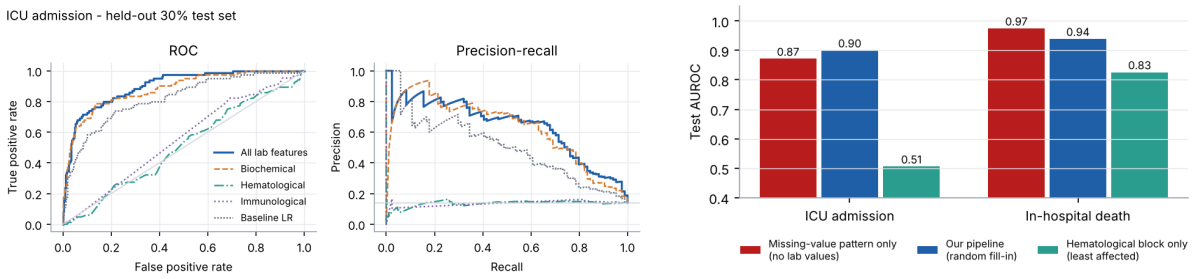


Figure: left, ROC and precision-recall for ICU admission; right, the missing-value-pattern diagnostic (test AUROC).

6. Can we do better? (improvement experiment)

We tried legitimate improvements, chosen with CV on the training split only and tested once: **B** six clinical ratio features (NLR, PLR, AST/ALT, urea/creatinine, LDH/albumin, CRP/albumin); **C** randomized-search-tuned LightGBM (40 draws); **D** C averaged over 5 random imputations (cancels the noise of one random fill-in); **E** D plus RF, ExtraTrees and LR (rank average). **ICU:** nothing beat the baseline in CV (CV AUROC: baseline 0.909, B 0.908, C 0.895, D 0.902, E 0.881), so the baseline stays. **Death:** D won in CV (AUPRC 0.79 to 0.85) and on the test set (AUROC 0.94 to 0.94, AUPRC 0.71 to 0.78; better in 100% of bootstrap resamples on AUPRC, 52% on AUROC), so it is the final death model (sensitivity 0.79, specificity 0.94, precision 0.57). Plain accuracy is a poor yardstick here: always answering "no" scores 86% (ICU) and 91% (death). We deliberately did not let models read the missing-value pattern (a NaN-aware LightGBM reaches AUROC 0.95 / 0.99) because that mostly reflects how the dataset was filled in, not patient physiology. The tables and curves in section 5 show the baseline pipeline; the test set was used for it and for the one improved model per task.

7. Conclusions

A lab-only model separates ICU-bound and fatal COVID-19 cases well on held-out data (ICU AUROC 0.90, death 0.94), echoing the paper's finding that fusing more data types helps (all blocks together beat any single block). Gradient boosting was best; FPR feature selection helped the ICU model (CV AUROC 0.91 vs 0.88 without selection) while the death model's best configuration uses no selection (CV AUROC 0.96; LASSO/FPR variants are within 0.01). **Caveats:** the dataset's imputation artifacts mean absolute numbers are probably optimistic - the least-affected hematological-only death model scores 0.83 - and ICU performance is carried by biochemistry; no external validation cohort, CT radiomics or ventilation label was available, so the paper's AUROCs (0.91 ICU, 0.92 ventilation, 0.83 death) are not directly comparable. With data-quality-safe collection (patient IDs, no label-dependent imputation) the same pipeline could be rerun unchanged. Future work: add radiomics features, nested CV with hyperparameter search, calibration and external validation.

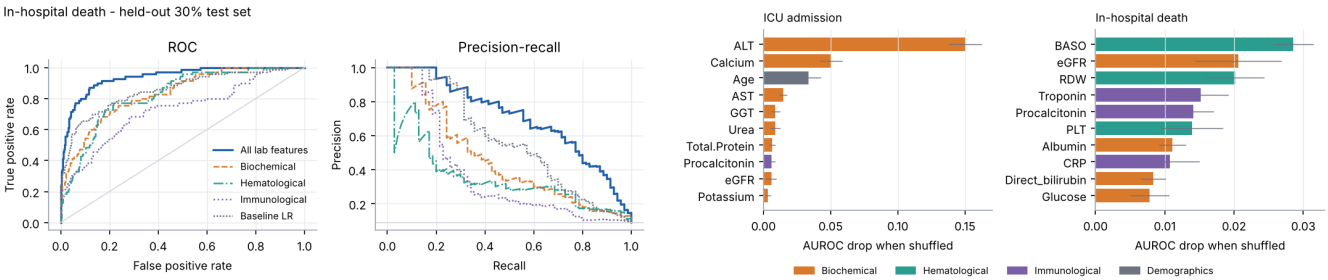


Figure: left, ROC and precision-recall for death; right, permutation importance of the top 10 features for both tasks (colour = data block).

References. Zhan & Li, CS229 Spring 2020, Stanford (Problem 12); Huyut, M.T., Routine Blood Values Datasets, Mendeley Data v1, doi:10.17632/w9mdp9k472.1; Batista et al. 2003 (SMOTEENN); Tibshirani 1996 (LASSO); Ke et al. 2017 (LightGBM); Pedregosa et al. 2011 (scikit-learn).